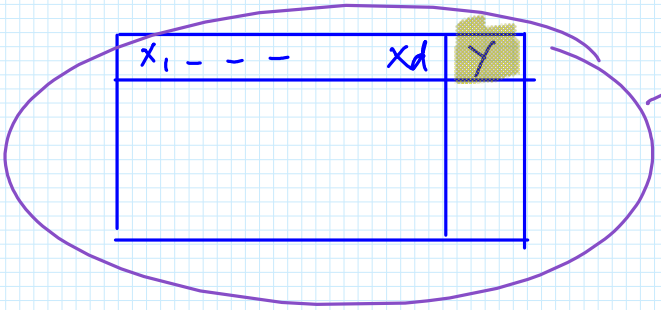
Boosting - 2

27 October 2025 21:12

Bagging → Variance [Tr & Te]

Boosting → Your Training error is high or traing accuracy is low



(Markov Model)
[Log Reg] → (62% accuracy)
(135 million rows)
→ Boosting → 79%
→ Boosting [Recall] → 32%

Agenda →
- Revise Boosting (G.B)
- Research Paper (Pseudo Code)
- Bias & Variance Analysis
- Advance techniques
- [Stochastic GBDT] {R.F + Boosting}
- EMG Signal Clf

Internals of GBDT

Let's understand the **pseudo code**

Input: training set $\{(x_i, y_i)\}_{i=1}^{n}$, a differentiable loss function $L(y, F(x))$, number of iterations M.

Algorithm:

1. Initialize model with a constant value:
$$F_0(x) = \arg\min_{\gamma} \sum_{i=1}^{n} L(y_i, \gamma).$$
2. For $m = 1$ to M:
   1. Compute so-called *pseudo-residuals*:
   $$r_{im} = -\left[\frac{\partial L(y_i, F(x_i))}{\partial F(x_i)}\right]_{F(x)=F_{m-1}(x)} \quad \text{for } i = 1, \ldots, n.$$
   2. Fit a base learner (or weak learner, e.g. tree) closed under scaling $h_m(x)$ to pseudo-residuals, i.e. train it using the training set $\{(x_i, r_{im})\}_{i=1}^{n}$
   3. Compute multiplier γ_m by solving the following one-dimensional optimization problem:
   $$\gamma_m = \arg\min_{\gamma} \sum_{i=1}^{n} L\left(y_i, F_{m-1}(x_i) + \gamma h_m(x_i)\right).$$
   4. Update the model:
   $$F_m(x) = F_{m-1}(x) + \gamma_m h_m(x).$$
3. Output $F_M(x)$.



* [1. Initialize model with a constant value: $F_0(x) = \arg\min_{\gamma} \sum_{i=1}^{n} L(y_i, \gamma)$.] = mean model

(Revise the last class quickly)

x_1 - - - x_d	y

→ F_0 = mean model = $\bar{y}$

$\text{Residual}_0 = (y - F_0)$

Fit a w.l to Residual_0

x_1 - - - x_d	Res_0

$h_1(x)$

Add this to final pred

$F_1 = F_0 + \gamma_1 h_1$

(current model after step 1)

* $\left[\frac{-\partial L}{\partial F_0}\right] = \text{P.R}$

$y - F_1 = \text{Residual}_1 \longrightarrow h_2(x) \longrightarrow F_2 = F_1 + \gamma_2 h_2$

$F_2 \longrightarrow F_3 \longrightarrow F_4$ - - F_M

$F_M = F_0 + \sum_{i=1}^{M} \gamma_i h_i$ (Final Model)

$F_M = F_0 + \sum_{i=1} \gamma_i h_i$ (Final ...)

Change (Residual) ⟶ Pseudo Residual

$\left(-\frac{\partial L}{\partial F_{M-1}}\right) = P.R$

STEP 1

Algorithms:

- Initialize model with a constant value :

$$F_0(x) = \underset{\gamma}{argmin} \sum_{i=1}^{n} L(y_i, \gamma)$$

mean model

We start by creating **STAGE 0** model i.e mean model.

- But that's not written in STEP - 1.



Total Loss $= \sum (y_i - \gamma)^2$

$\gamma = \frac{\sum y_i}{N}$ minimise loss function

$ax^2 + bx + c = F(x)$

$2ax + b = F'(x) = 0$

$x = -b/2a$

$L = \sum_{i=1}^{N} (y_i - \gamma)^2$

$\frac{\partial L}{\partial \gamma} = \sum_{i=1}^{N} 2(y_i - \gamma)(-1) = 0$

$\frac{\partial L}{\partial \gamma} = 0$

$$\frac{\partial}{\partial \gamma} \sum_{i=1}^{N} (y_i - \gamma) = 0$$

$$\Rightarrow (y_1 - \gamma) + (y_2 - \gamma) + \cdots (y_N - \gamma) = 0$$

$$\Rightarrow \gamma = \left(\frac{y_1 + \cdots y_N}{N}\right) = \bar{y}$$

What does STEP 1 mean ?

–For a given loss $L(y^i, \gamma)$,

find γ which minimizes the loss $L(y^i, \gamma)$



Which value of γ minimizes square loss ?

To find minima, we take derivative of loss function

$$F_0(x) = \underset{\gamma}{argmin} \sum_{i=1}^{n} L(y_i, \gamma)$$

$$F_0(x) = \underset{\gamma}{argmin} \sum_{i=1}^{n} (y_i - \gamma)^2$$

$$\frac{\partial L}{\partial \gamma} = \frac{\partial \sum_{i=1}^{n} (y_i - \gamma)^2}{\partial \gamma}$$

$$\frac{\partial L}{\partial \gamma} = \sum_{i=1}^{n} -2(y_i - \gamma)$$

To find minima, equate it to 0

$$\frac{\partial L}{\partial \gamma} = \sum_{i=1}^{n} -2(y_i - \gamma)$$

$$\frac{\partial L}{\partial \gamma} = \sum_{i=1}^{n} (y_i - \gamma) = 0$$

Taking, γ out of summation,

$$\frac{\partial L}{\partial \gamma} = \sum_{i=1}^{n} y_i - n\gamma = 0$$

$$\frac{\partial L}{\partial \gamma} = \sum_{i=1}^{n} y_i = n\gamma$$



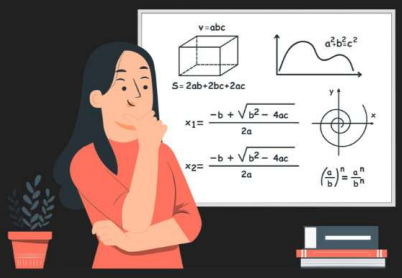
$$\gamma = \frac{1}{n} \sum_{i=1}^{n} y_i$$ → Mean of all y_i

Notice That:

- γ Which minimizes square loss is nothing but mean value
- The complex equation is simply telling us to create mean model

After finding value of M_0 (Stage 0)

- We need to find $h_1(x)\ ,\ h_2(x)\ .....\ h_m(x)$
- As well as weights $\gamma_1\ ,\ \gamma_2\ ,\ \gamma_3\ ,\ .....\gamma_m$

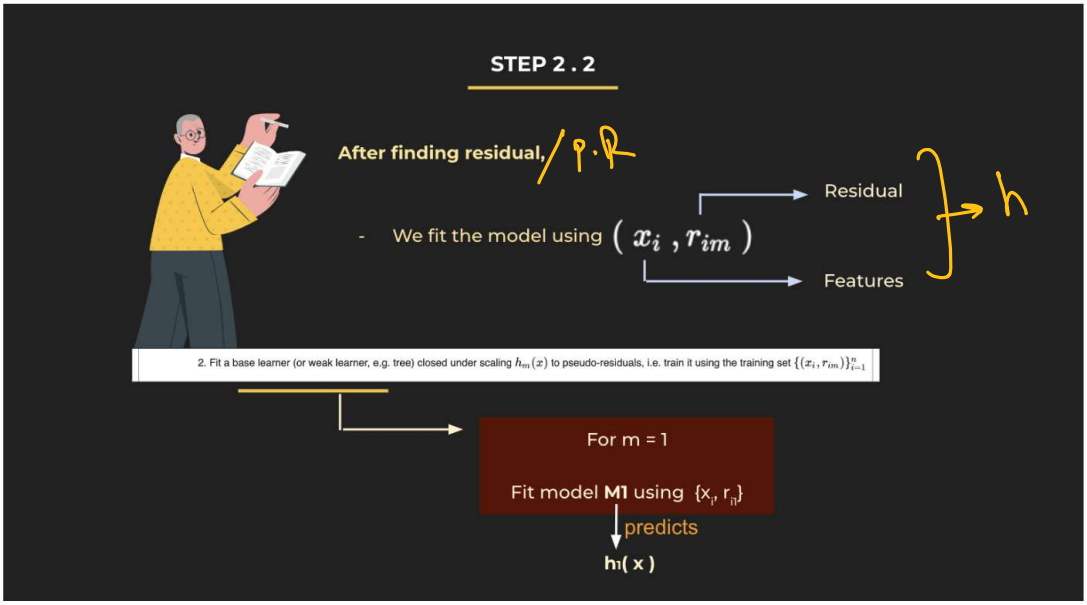


$$F_M = F_0 + \gamma_1 h_1 + \gamma_2 h_2 + \cdots + \gamma_M h_M$$

STEP 2

Create M models (Stage 1 to Stage M)

2. For $m = 1$ to M:
   1. Compute so-called *pseudo-residuals*:
      $$r_{im} = -\left[\frac{\partial L(y_i, F(x_i))}{\partial F(x_i)}\right]_{F(x)=F_{m-1}(x)} \quad \text{for } i = 1, \ldots, n.$$
      [g.d in functional space]
   2. Fit a base learner (or weak learner, e.g. tree) closed under scaling $h_m(x)$ to pseudo-residuals, i.e. train it using the training set $\{(x_i, r_{im})\}_{i=1}^n$
   3. Compute multiplier γ_m by solving the following one-dimensional optimization problem:
      $$\gamma_m = \arg\min_{\gamma} \sum_{i=1}^{n} L\left(y_i, F_{m-1}(x_i) + \gamma h_m(x_i)\right).$$
   4. Update the model:
      $$F_m(x) = F_{m-1}(x) + \gamma_m h_m(x).$$



model of (x_i, γ_i)

step 3

$F_3 = F_2 + \gamma_M h_3$ (To avoid overfitting)

Mom → h_{mom}

Civil Engg Uncl → h_{uncl}

Sciehst Dad → h_{dad}

(Take Civil Engg or Not) = Not $[h_{mom} + h_{uncl} + h_{dad}]$

No + $[0.01(\quad) + 0.9(\quad)]$

$+ 0.09(h_{dad})$

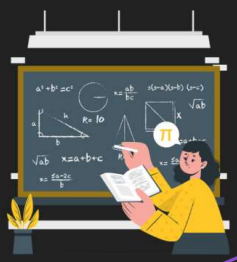
How to find multiplier (γ_m)

After learning h1 (x) - We need to find γ

STEP 2 . 3

3. Compute multiplier γ_m by solving the following one-dimensional optimization problem:

$$\gamma_m = \arg\min_{\gamma} \sum_{i=1}^{n} L\left(y_i, F_{m-1}(x_i) + \gamma h_m(x_i)\right).$$



In simple terms, this equation is saying ;

- To find γ which minimizes given **loss**

$$L(y_i, F_{m-1}(x_i) + \gamma h_m(x_i))$$

y_i : Actual Value; $F_{m-1}(x_i) + \gamma h_m(x_i)$: $\hat{y}$ (Predicted Value)

$L(y, \hat{y}) \Rightarrow$ $L(y, F_{M-1} + \gamma h_m)$ ← mth step

greedy decision → (already known)

F_1 ← γ_1

$L(y, \hat{y} + \gamma_1 h_1)$ → known

(1D optim eq^n)

say $\left((2-\gamma)^2 + 3\gamma - 8\gamma^3 + 10\gamma^2 \right) = \text{Loss}$

↓

γ which minimises the loss

$2(2-\gamma)(-1) + 3 - 24\gamma^2 + 20\gamma = 0$ $\left(\frac{d\,\text{Loss}}{d\gamma} = 0\right)$

We keep doing sub-steps for M iteration i.e finding — $h_1(x), h_2(x), h_3(x) \ldots h_m(x)$; $\gamma_1, \gamma_2, \gamma_3 \ldots \gamma_m$

Finally, we use these to make final model $F_m(x)$

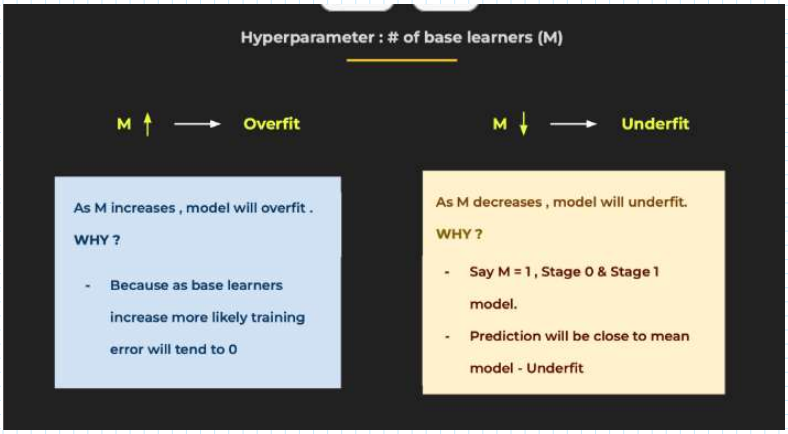
3. Output $F_M(x)$.

$$F_m(x) = h_0(x) + \gamma_1 h_1(x) + \gamma_2 h_2(x) + \ldots + \gamma_m h_m(x)$$

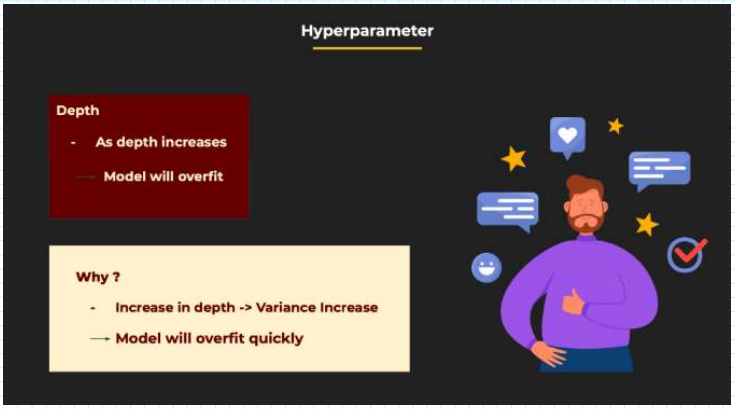
How to define / find M →

How to define / find

Effect of M (Bias - Variance Trade off)



(Base Learner) → stump / D. Tree



22 : 14 *

(RF) + (Boosting)

(Applied) RF → magically Not overfits

[Trees → independent] (RS+CS)

Boosting → (overfit a lot)

Regularization by Shrinkage

Final Model equation is : $F_m(x) = h_0(x) + \sum_{m=1}^{M} \gamma_m . h_m(x)$

To regularize, we add an regularization term i.e learning rate

$$F_m(x) = h_0(x) + \nu \sum_{m=1}^{M} \gamma_m . h_m(x)$$

ν : Learning Rate

Range : $o \leq \nu \leq 1$

Notes :

- Adding learning rate is reducing the impact of Mth .

  Hence, reducing overfit.

< 0.1 work well

Ans 1

What happens if we have learning rate as a very small value ?

4 options

Active Duration(Most preferred: 30 seconds)

Appears for 45 Secs

A	It will take less iterations (M) for model to converge
B	It will take more iterations (M) for model to converge
C	No impact. Model will converge as it was before introduction of learning rate
D	None of the above

$F_3 = \bar{F}_2 + (1)\gamma_2 h_2$

$F_3^1 = \bar{F}_2 + 0.01\ \gamma_2 h_2$

Blindfold

$3(0.1) = 0.3$

(Random)

Stochastic Gradient Boosting

PROBLEM : GBDT overfit a lot

(High Variance) → sol^n ? * (Bagging)

How to reduce variance ? - **Randomization** → Row sampling + column sampling

- Can use the some concept of randomization (as used in RF) to reduce variance

- This variation of GBDT is called **'Stochastic Gradient Boosting'** [Stochastic GB]

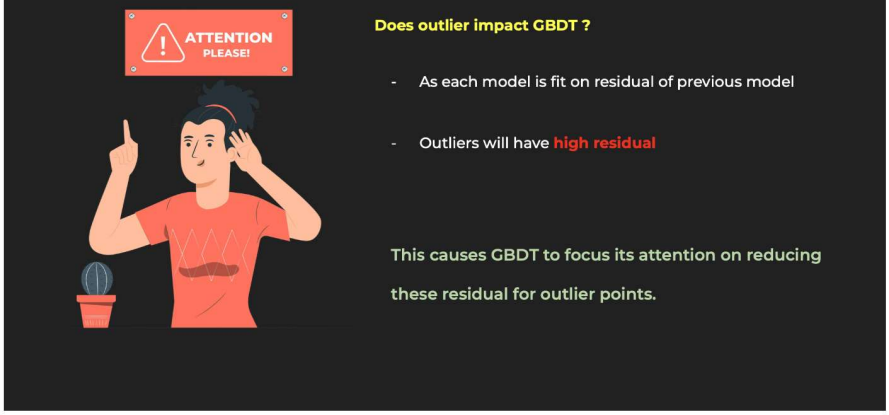
GBDT → **Pseudo Residual** + **Additive Combing**

Stochastic GDBT → **GBDT** + **Randomization**

Row Sampling + Column Sampling

- skLearn provides ability of row samping
  - Using **subsample** hyperparameter
  - And column sampling using **max_features** hyperparameter

Ques Does Outlier impact GBDT ?



Does outlier impact GBDT ?

- As each model is fit on residual of previous model
- Outliers will have **high residual**

This causes GBDT to focus its attention on reducing these residual for outlier points.

Hint : I can diminish the effect by updating the Loss fn

L1 MSE & L2 MAE

$\Sigma (y - \hat{y})^2$ $\Sigma |y - \hat{y}|$

Power of Boosting

→ Any Loss fn

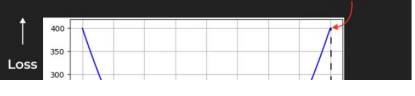
→ Residual

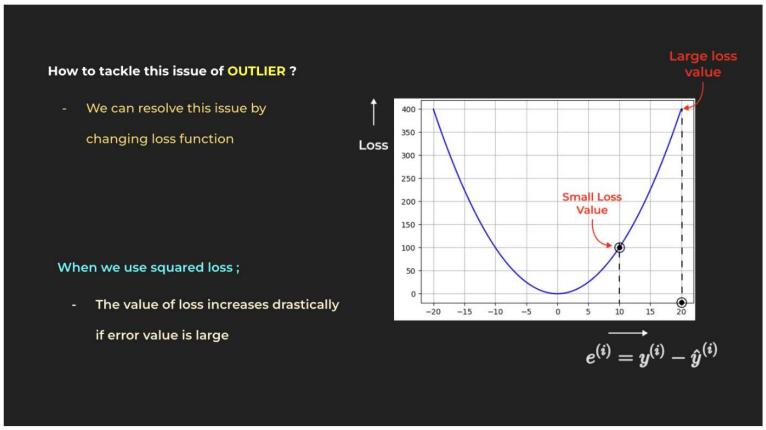
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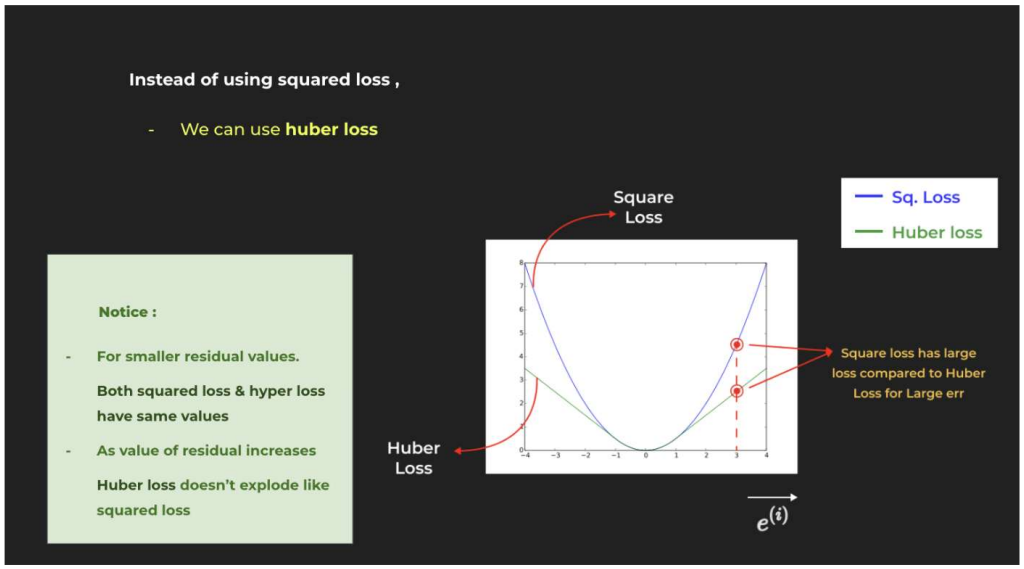
Pseudo Residual

How to tackle this issue of OUTLIER ?

- We can resolve this issue by changing loss function





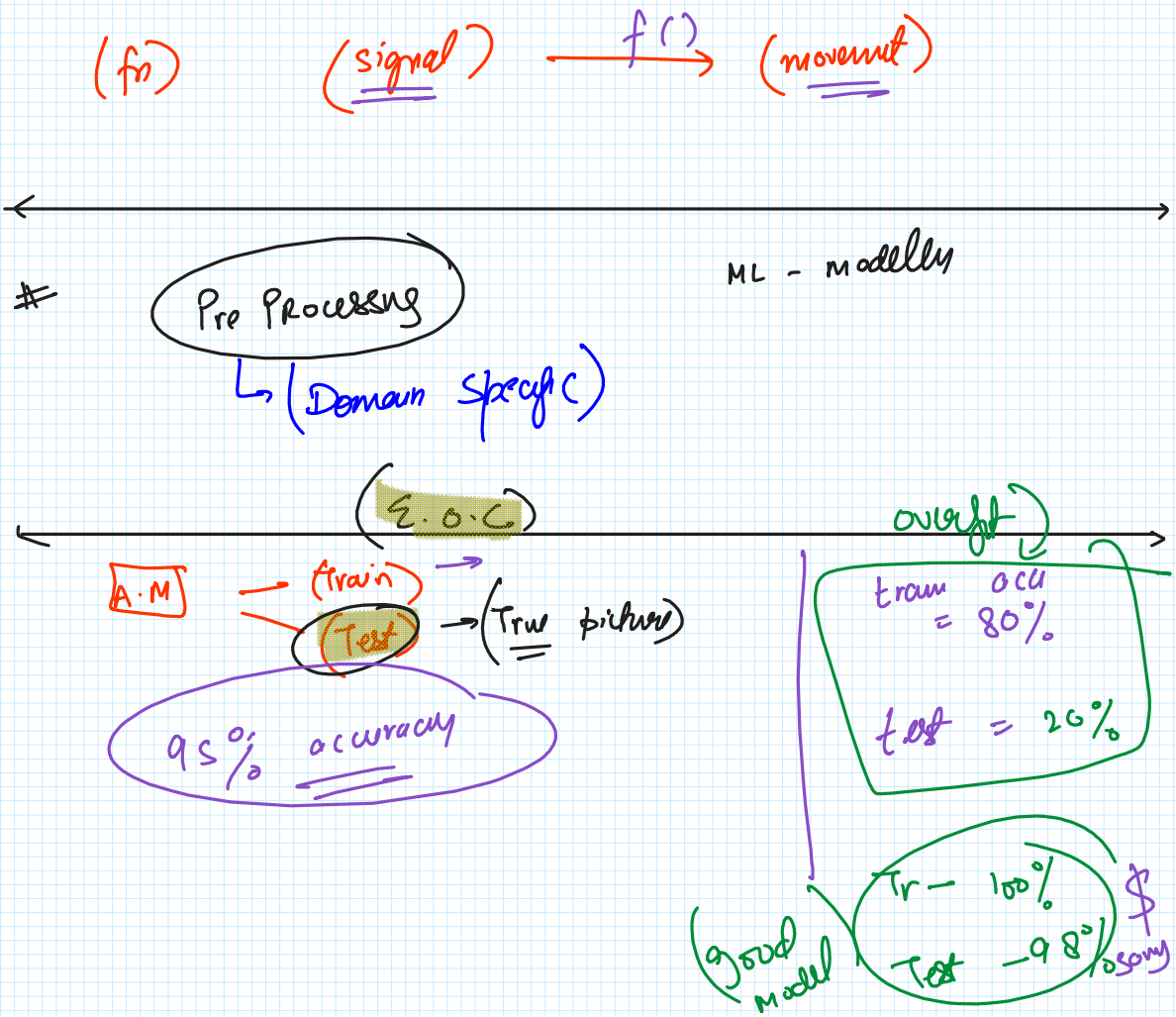


(#) Project (EMG) → * [Electromiography]

- Technique to study electrical signals produced by muscular movement. → EMG

(Problem statement) → (Prosthetic Arm) (Fall)




(fn)
(signal)
f()
(movement)
ML - modelly
Pre Processing
(Domain Specific)
E.O.C
overfit
A.M
(train)
Test
(True picture)
95% accuracy
train acu = 80%
test = 26%
Tr - 100%
Test - 98%
good model
$
sorry